

Work, Energy & Power

Core Principles of Energy

Energy is the capacity to do work. It is a scalar quantity measured in joules (J).

The Principle of Conservation of Energy states that:

“Energy cannot be created or destroyed. It can only be transferred from one form to another or from one object to another. The total energy in a closed system remains constant.”

Forms of Energy

- **Kinetic Energy (K.E.):** Energy of an object due to its motion.
- **Gravitational Potential Energy (G.P.E.):** The energy stored when a mass is moved within a gravitational field, due to work done in a gravitational field forces.
- **Elastic Potential Energy (E.P.E.):** energy stored in an object due to changing its shape (deformation)
- **Thermal/Internal Energy:** The sum of kinetic and potential energies of the atoms/molecules within a substance.
- **Electrical Potential Energy:** The energy stored when a charge is moved within an electric field, due to work done in an electric field.
- **Chemical Energy:** Energy stored in the bonds between atoms, released in chemical reactions.
- **Nuclear Energy:** Energy stored within the nucleus of an atom.

Energy Conversions & Efficiency

- Energy is constantly being converted from one form to another (e.g., a battery converts chemical energy to electrical energy).
- In any process, some energy is **usefully transferred** and some is **wasted** (often dissipated as thermal energy to the surroundings).
- **Examiner Tip:** The "wasted" energy isn't lost from the universe (Conservation of Energy), but it becomes unavailable to do the desired useful work for the system in question.

Work Done

Definition & Equation

Work done is defined as force $\times$ displacement in the direction of the force.

$$W = Fs \cos \theta$$

- W = work done (J)
- F = magnitude of the force (N)
- s = magnitude of the displacement (m)
- θ = angle between the force vector and the displacement vector.

SI Unit: joule (J). $1J = 1Nm$

Interpretation of Work

- **Work as Energy Transfer:** *Work done on a system equals the energy transferred to it.*
- **Positive Work ($\theta < 90^\circ$):** Force has a component in the direction of motion. Energy is transferred **to** the object (e.g., a force accelerating a car).
- **Negative Work ($\theta > 90^\circ$):** Force has a component opposite to motion. Energy is transferred **from** the object (e.g., friction slowing a moving object).
- **Zero Work ($\theta = 90^\circ$):** Force is perpendicular to displacement. **No** energy is transferred in the direction of the force (e.g., the tension in a string for an object in uniform circular motion, or the normal contact force on a horizontally moving object).

Work Done Against Gravity & Friction

- **Against Gravity:** $W = mgh$, where h is the **vertical** height moved.
- **Against Constant Friction:** $W = F_{friction} \times s$. This work is dissipated as thermal energy.

Gravitational Potential Energy (G.P.E.)

Derivation & Equation

Consider lifting a mass m at constant velocity to a height h . The upward force needed equals its weight, mg . Work done = Force $\times$ distance.

$$W = F \times s = (mg) \times h$$

This work done is stored as G.P.E.:

$$\Delta E_p = mg\Delta h$$

- ΔE_p = change in G.P.E. (J)
- m = mass (kg)
- g = gravitational field strength (Nkg^{-1} or Nkg^{-1})
- Δh = change in vertical height (m)

Important: h is measured from a reference **zero point**, in other word there is no specific point/level that represents zero . Only *changes* in G.P.E. are physically meaningful.

Kinetic Energy (K.E.)

Derivation & Equation

Consider a constant resultant force F accelerating a mass m from rest ($u=0$) over displacement s , reaching velocity v .

1. Work done by force: $W = Fs$.
2. From Newton's 2nd Law: $F = ma$.
3. From kinematics: $v^2 = u^2 + 2as \Rightarrow s = \left(\frac{v^2 - u^2}{2a}\right)$.

Substitute (2) and (3) into (1):

$$W = (ma) \times \left(\frac{v^2 - u^2}{2a}\right) = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$\text{If } u = 0 \Rightarrow E_k = \frac{1}{2}mv^2$$

Elastic Potential Energy (E.P.E.)

Hooke's Law & Derivation

For a spring (or wire) within its limit of proportionality:

$$F = kx$$

- F = restoring force (N)
- k = spring constant (stiffness) (Nm^{-1})
- x = extension (or compression) from natural length (m)

The work done in stretching the spring is stored as E.P.E. This equals the **area under the Force-Extension graph**.

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}fx = \frac{1}{2} \times x \times (kx) = \frac{1}{2}kx^2$$

Examiner Tip: This equation is **only valid** if it has **not exceeded its limit of proportionality**, or in other words Hooke's Law is obeyed (the graph is a straight line through the origin).

A Falling Object

Without Air Resistance (Conserved System)

- System: Object + Earth.
- Energy is converted from G.P.E. to K.E.

$$\text{Loss in G.P.E.} = \text{Gain in K.E.}$$

- **Total mechanical energy (G.P.E. + K.E.) is constant.**

With Air Resistance (Non-Conserved System)

- System: Object + Earth + Air.
- Some G.P.E. is converted to thermal energy of the object and air.

$$\text{Loss in G.P.E.} = \text{Gain in K.E.} + \text{Work done against air resistance}$$

- The object hits the ground with a **lower speed** than in the frictionless case.
- **On Impact:** Remaining K.E. is transferred to do work deforming the ground/object, and into sound and thermal energy. Mechanical energy is **not conserved**.

Power

Definition & Equations

Power is work done per unit time.

$$P = \frac{W}{t} = \frac{E}{t}$$

SI Unit: watt (W). $1\text{W} = 1\text{ J s}^{-1}$

Derivation of $P = Fv$

For a constant force F moving an object at constant velocity v :

$$P = \frac{W}{t} = \frac{F \times s}{t}$$

Since $v = \frac{s}{t}$:

$$P = Fv$$

- Here, F is the force **in the direction of motion** (e.g., the driving force of an engine).
- **Crucially Applicable:** When an object moves at constant velocity, the net force is zero. F in $P = Fv$ is the driving force, which is equal in magnitude (and opposite in direction) to the total resistive force.
- Power is the gradient of a energy-time graph.

Efficiency

Definition & Equation

Efficiency measures the proportion of energy (or power) input that is usefully transferred.

$$Efficiency = \frac{\text{Useful Energy Output}}{\text{Total Energy Input}} \times 100\%$$

Or

$$Efficiency = \frac{\text{Useful Power Output}}{\text{Total Power Input}} \times 100\%$$

Efficiency is a dimensionless ratio or percentage, always **< 100%** in real systems due to energy dissipation.

Problem-Solving Framework

1. **Define the System:** Clearly identify the object(s) you are considering. Is it isolated?
2. **Identify Energy Types:** Note all forms of energy at the start (initial) and end (final) of the process. Consider G.P.E., K.E., E.P.E., thermal.

3. **Apply Energy Principles:**

- If no resistive forces act $\Rightarrow$ **Conservation of Mechanical Energy:**

$$E_{p,initial} + E_{k,initial} = E_{p,final} + E_{k,final}$$

- If resistive forces (friction, air resistance) act $\Rightarrow$ **Work-Energy Theorem:**

$$\Delta E_k + \Delta E_p = W_{non-conservative}$$

“The change in mechanical energy equals the work done by non-conservative forces”

4. **Solve:** Substitute equations and solve for the unknown. Check units and reasonableness of answer.